

VAIBHAV JATHAR

Pune, Maharashtra

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Education

Motilal Nehru National Institute Of Technology.

Bachelor of Technology in Computer Science and Engineering.

Prayagraj.

Expected May 2026

Experience

Rises Analytics Solutions Pvt Ltd

May 2026 – Present

Applied AI Intern

(Remote) - Pune, Maharashtra.

- Architected a full-stack **Clinical Decision Support System (CDSS)** integrating a **PyTorch DenseNet121** model with FastAPI and React, enabling multi-label detection across **14 respiratory pathologies** with confidence-based triage.
- Built a custom **OHIF v3 Viewer** extension to process **DICOM images**, leveraging **Groq LLM (LLaMA3)** to auto-generate structured, clinical-grade **TRAP (Triage, Risk, Assessment, Plan) reports** for accelerated radiology workflows.
- Engineered** a real-time diagnostic workflow, integrating the **Web Speech API** for seamless voice dictation and automating one-click DICOM Structured Report (SR) generation to ensure secure local patient data persistence.

Savla Associates

June 2025 – July 2025

Software Developer Intern

(Remote) - Pune, Maharashtra.

- Innovated and deployed the firm's first responsive corporate website using Next.js and Tailwind CSS, driving an estimated **30% increase in inbound client inquiries** and establishing a modern digital presence.
- Spearheaded the development and deployment of a life insurance calculator using Firebase; automated policy quotes, which saved senior associates **6+ hours weekly** in consultation time.
- Collaborated with senior associates in an Agile environment to gather project requirements and translate business needs into functional, user-centric web features.

Projects

IoT Wearable Function Name Predictor 🧠 | Python, Scikit-learn, SciPy, NLP

April 2026

- Developed a **100% offline edge AI intent classifier** mapping natural language commands to IoT SDK functions, achieving **99.99% test accuracy** and sub-millisecond latency.
- Collaborated a **Hierarchical Logistic Regression router** utilizing dual word-and-character TF-IDF fusion, strictly confining the final model artifact to **860 KB for memory-constrained microcontrollers**.
- Designed an **Out-of-Distribution (OOD) Reject Gate** leveraging Shannon entropy and Top-1/Top-2 confidence gaps, guaranteeing **100% OOD detection** and preventing catastrophic hardware misfires.
- Enhanced **transcription fault tolerance** by integrating **3-to-6 character n-grams**, absorbing voice-to-text typos and **boosting baseline accuracy by 30%**.

Enterprise AI Support Orchestrator 🧠 | Live Infographics 📊 | FastAPI, LangGraph, FAISS, Streamlit

Mar 2026

- Orchestrated an autonomous **7-node LangGraph DAG** agent with **Groq (Llama-3.1)**, slashing support triage time from 24 hours to **<5 seconds** with **92% accuracy**.
- Applied a fault-tolerant **FAISS RAG** pipeline with a graceful degradation fallback, guaranteeing **100% uptime** for semantic knowledge retrieval.
- Created a resilient **FastAPI & SQLAlchemy** backend, utilizing custom **Pydantic** schemas to strictly eliminate LLM JSON hallucinations and enforce type-safety.
- Deployed a **Streamlit** UI with **Plotly** analytics for Human-In-The-Loop escalation, accelerating the build cycle **3x** via **Claude Code** pair-programming.

Certifications

DeepLearning.AI & Coursera: Design, Develop, and Deploy Multi-Agent Systems with CrewAI 🧠 (Feb 2026)

Technical Skills

- Languages:** Python, TypeScript, JavaScript (ES6+), C++, SQL (PostgreSQL), HTML/CSS.
- Frameworks & Libraries:** PyTorch (TorchXRayVision), FastAPI, React.js / Next.js, LangChain / LangGraph, Scikit-Learn, Pandas / NumPy, pydicom, Tailwind CSS.
- Dev-Tools & Platforms:** Docker, Git/GitHub, Vercel / Render, FAISS, Groq API, Firebase, OHIF Viewer v3
- Domain Expertise:** Deep Learning, Machine Learning, Large Language Models (LLMs), RAG & Agentic AI, NLP, Edge AI, Medical Imaging Standards (DICOM, PACS)